Modern Algebra 1

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Problem 1

Proof. Let R be a finite, nontrivial, commutative ring with no zero divisors. Let $a \in R^*$, and let $\lambda_a : R^* \to R^*$ be the map $b \mapsto ab$. We want to show $a \in \lambda(R^*)$. This implies there exists some $k \in R$ such that $ak = a$. This is equivalent to showing λ is surjective, and since R is finite, this is equivalent to showing λ is injective.

Suppose $\lambda_a(b) = \lambda_a(c)$, implying that $ab = ac$. It follows that $a(b - c) = 0$. Since R has no zero divisors and $a \neq 0$, we have $b - c = 0$. It follows that $b = c$, and λ_a is injective. Therefore, for all $a \in R^*$, there exists $k_a \in R$ such that $ak_a = k_a a = a$. We will now show this k is unique.

Let $a, b \in R^*$ such that $ak_a = a$ and $bk_b = b$. It follows since R is commutative that

$$ab = ak_a bk_b = abk_a k_b.$$

It follows that $a\left(b - bk_a k_b\right) = 0$. Since $a \neq 0$ and R has no zero divisors, $b = bk_a k_b$. Since $bk_b = b$ and R is commutative, it follows that $b = bk_a$. Since $k_a 0 = 0$, we know $rk_a = r$ for any arbitrary $a \in R^*$ and $r \in R$. Therefore, k_a is an identity in R for any a. Since the identity is unique in R, we know $k_a = k_b$ for all $a, b \in R$. Therefore, R has a unity. ■

Problem 2

Part (a). Let $I, J \subset R$ be ideals of R. First, we show $I \cap J$ is a subring of R by the subring test. Let $r, s \in I \cap J$. Therefore, $r, s \in I, J$. Since I is a subring, $r - s \in I$ since I is closed under addition and has inverses. Since J is a subring, $r - s \in J$ by the same logic. Therefore, $r - s \in I \cap J$. Now notice that $rs \in I$ since I is closed under multiplication, and by the same logic $rs \in J$. Therefore, $rs \in I \cap J$, and $I \cap J$ is a subring. Finally, let $a \in R$. Notice that $ar \in I$ since I is an ideal, and $ar \in J$ because J is an ideal. Therefore, $ar \in I \cap J$, and $I \cap J$ is an ideal.

Part (b). Let $\phi : R \to S$ be a ring homomorphism. Let $r \in \ker \phi$. It follows that $\phi(r) = 0$. Let $a \in R$. It follows that $\phi(ar) = \phi(a)\phi(r) = \phi(a)0 = 0$. Therefore, $ar \in \ker \phi$. It remains to be shown $\ker \phi$ is a subring of R.

Let $r, s \in \ker \phi$. It follows that $\phi(r - s) = \phi(r) - \phi(s) = 0 - 0 = 0$. Therefore, $r - s \in \ker \phi$. Now notice that $\phi(rs) = \phi(r)\phi(s) = 0 \cdot 0 = 0$. Therefore, $rs \in \ker \phi$, and $\ker \phi$ is an ideal of R.

Part (c). Let $J \subseteq S$ be an ideal. First we show $\phi^{-1}(J)$ is a subring of R, which probably follows from properties of homomorphisms but at the time of doing this homework we haven't introduced homomorphisms yet.

Let $r, s \in \phi^{-1}(J)$. It follows that $\phi(r - s) = \phi(r) - \phi(s)$. Since $\phi(r), \phi(s) \in J$, which is closed under addition, $\phi(r) - \phi(s) \in J$. Now notice that $\phi(rs) = \phi(r)\phi(s) \in J$ by the same logic. Therefore, $\phi^{-1}(J)$ is a subring of R.

Now let $a \in R$. It follows that $\phi(ar) = \phi(a)\phi(r)$. Since J is an ideal and $\phi(r) \in J$, then $\phi(a)\phi(r) \in J$. Therefore, $ar \in \phi^{-1}(J)$, and $\phi^{-1}(J)$ is an ideal.

The converse is not true in general. Let I be an ideal of R and $\phi : R \to S$ a ring homomorphism. Let $r', s' \in \phi(I)$. By definition, there exist $r, s \in I$ such that $r' = \phi(r)$ and $s' = \phi(s)$. It follows that

$r' - s' = \phi(r - s) \in \phi(I)$, and similarly $r's' \in \phi(I)$. Therefore, $\phi(I)$ is a subring of S. However, let $a \in S$. If $a \in \phi(R)$, then for all $a' \in R$ such that $\phi(a') = a$, we know $a\phi(r) = \phi(a'r) \in \phi(I)$ because I is an ideal. However, if $a \notin \phi(R)$, then there is no guarentee that $a\phi(r) \in \phi(I)$. Therefore, the converse is not true in general, but is true for surjective homomorphisms. Moreover, the converse is true for the subset $\phi(R)$. ■

Problem 3

Notation. Let 0_I be the zero function $0_I(x) = 0$; let 1_I be the indicator function $1_I(x) = 1$; let $0, 1 \in \mathbb{R}$.

Part (a). Let $f \in \text{Fun}(I, \mathbb{R})$. The additive inverse of f in $\text{Fun}(I, \mathbb{R})$ is the pointwise additive inverse $-f(x)$. For any point $f(x) \in \mathbb{R}$, $-f(x) + f(x) = 0$. Therefore, $(f - f)(x) = 0$ for all x, and $f - f = 0_I$.

Part (b). The function $1_I : I \to \mathbb{R}$ defined as $1_I(x) = 1$ is the multiplicative identity. Let $f \in \text{Fun}(I, \mathbb{R})$. It follows that

$$(1_I f)(x) = 1_I(x) f(x) = 1 f(x) = f(x).$$

Note that this last step follows since $f(x) \in \mathbb{R}$, and 1 is the unity in $\mathbb{R}$.

Part (c). A function $f \in \text{Fun}(I, \mathbb{R})$ is a unit if and only if $0 \notin f(I)$. Suppose f has $0 \notin f(I)$. Then define $f^{-1} \in \text{Fun}(I, \mathbb{R})$ such that

$$f^{-1}(x) = \frac{1}{f(x)}.$$

It follows that

$$\left(f^{-1} f\right)(x) = f^{-1}(x) f(x) = \frac{1}{f(x)} f(x) = 1_I.$$

Now we show the converse by contrapositive. Let $0 \in f(I)$. It follows that there exists at least one $a \in I$ such that $f(a) = 0$. The identity has $1_I(a) = 1$ by definition, so for f^{-1} to exist, $f^{-1}(a)$ must be a number such that $f^{-1}(a)0 = 1$. Since $\mathbb{R}$ is a field and $f^{-1}(x) \in \mathbb{R}$, we know $f^{-1}(x)0 = 0 \neq 1$, and by contradiction f is not a unit.

Part (d). A function is a zero divisor if and only if it is not a unit. By the contrapositive of Homework 9 Problem 2, we already know if it is a zero divisor, it is not a unit; we only prove if is not a unit then it is a zero divisor. Let $f \in \text{Fun}(I, \mathbb{R})$ not a unit. Then by (c), $0 \in f(I)$. Let A be the set of all $a \in I$ such that $f(a) = 0$. Let $\lambda \in \mathbb{R} \setminus \{0\}$, and define $g_\lambda \in \text{Fun}(I, \mathbb{R})$ as the map

$$g_\lambda(x) = \begin{cases} \lambda, & x \in A \\ 0, & x \notin A \end{cases}.$$

Notice that

$$(fg)(x) = f(x)g(x) = \begin{cases} 0\lambda = 0, & x \in A \\ f(x)0, & x \notin A \end{cases} = 0.$$

Therefore, since $g \neq 0_I$, f is a zero divisor.

Part (e). First, we show J is a subring of R. Let $f, g \in J$. It follows that

$$(f - g)\left(\frac{1}{2}\right) = 0 - 0 = 0, \qquad (f - g)\left(\frac{1}{3}\right) = 0 - 0 = 0.$$

Similarly, $(fg)(1/2) = (fg)(1/3) = 0$. Therefore, J is a subring. Now let $\phi \in \text{Fun}(I, \mathbb{R})$. It follows that

$$(\phi f)\left(\frac{1}{2}\right) = \phi\left(\frac{1}{2}\right) 0 = 0.$$

Additionally,

$$(\phi f)\left(\frac{1}{3}\right) = \phi\left(\frac{1}{3}\right) 0 = 0.$$

Therefore, $\phi f \in J$. Therefore, J is an ideal of $\text{Fun}(I, \mathbb{R})$.
Part (f). I will quickly prove that for all quotient rings R/I for R a commutative ring with unity, R/I is commutative and has unity. This proof will be referenced in problem 5.

Proposition 1. *Let R be a commutative ring with unity, and let I be an ideal of R. Then R/I is a commutative ring with unity.*

Proof. Let $a + I, b + I \in R/I$. Since $ab = ba$, it follows that

$$(a + I)(b + I) = ab + I = ba + I = (b + I)(a + I).$$

Notice that $1 + I \in R/I$. For any $a + I \in R/I$, we have

$$(1 + I)(a + I) = 1a + I = a + I.$$

Therefore, a unity exists. ■

Using this fact, we know that $\text{Fun}(I, \mathbb{R})/J$ is a commtuative ring with unity. We will characterize all zero divisors in $\text{Fun}(I, \mathbb{R})$. Since $J = 0$ in this ring, any $f, g \in \text{Fun}(I, \mathbb{R})$ satisfying $fg + J = J$ are zero divisors. Notice that if either $f \in J$ or $g \in J$, then $fg \in J$ since J is an ideal. Therefore, any element of J is a zero divisor, which is expected since for all $f \in J$, we have $f + J = J = 0$.

Now let $f, g \notin J$. Then f, g are zero divisors if and only if one of the following is satisfied:

1. $$f\left(\frac{1}{2}\right) = 0, \qquad g\left(\frac{1}{3}\right) = 0.$$

2. $$f\left(\frac{1}{3}\right) = 0, \qquad g\left(\frac{1}{2}\right) = 0.$$

Problem 4

Proof. Let S be a field. Let I be an ideal such that there exists some $a \in I$ such that $a \neq 0$. It suffices to show $I = S$. The existence of such an ideal is guarenteed by the fact that at least two distinct trivial ideals exist for $\{0\} \neq S$. Since $a \in I$ and $a \neq 0$, there exists $a^{-1} \in S$ such that $a^{-1}a = 1 \in I$. Since $1 \in I$, it follows that $I = S$. Therefore, any ideal $I \neq \{0\}$ implies $I = S$.

Now we show the converse. Let S be a ring with maximal idea $\{0\}$. It follows that the only ideals are the trivial ideals. Since S has unity, for all $a \in S \setminus \{0\}$, there exists $1a \in \langle a \rangle$. Since $\langle a \rangle$ is an ideal of S and the only ideals are $\{0\}$ and R, we have $\langle a \rangle = S$ for all $a \in S \setminus \{0\}$. Since $\langle a \rangle = S$, we have $1 \in \langle a \rangle$. Therefore, there exists some $a^{-1} \in S$ such that $aa^{-1} = 1$ for all $a \in S \setminus \{0\}$. Therefore, all elements of S are units. Since all units are not zero divisors, S is a field. ■

Problem 5

Part (a). Let I be an ideal of R, let $\mathcal{J}$ be the set of all ideals of R containing I, and let $\mathcal{J}'$ be the set of all ideals of R/I. We will show the map $\phi : \mathcal{J} \to \mathcal{J}'$ defined as $J \mapsto \pi(J)$ is a bijection.

First, we will show ϕ is well-defined by showing $\pi(J)$ is an ideal in R/I. Let $\alpha + I \in \pi(J)$ and $r + I \in R/I$. It follows that $\alpha \in J$ and $r \in R$. Since J is an ideal, $\alpha r \in J$, and thus

$$(\alpha + I)(r + I) = \alpha r + I \in \pi(J).$$

It remains to be shown $\pi(J)$ is a ring. Let $\alpha + I, \beta + I \in \pi(J)$. It follows that $\alpha, \beta \in J$, and thus $\alpha - \beta, \alpha\beta \in J$. We then have

$$(\alpha + I) - (\beta + I) = (\alpha - \beta) + I \in \pi(J).$$

$$(\alpha + I)(\beta + I) = \alpha\beta + I \in \pi(J).$$

Therefore, $\pi(J)$ is an ideal of R/I. This suffices to show $\phi(J)$ is well-defined. Now we will show ϕ is surjective. Let $J' \in \mathcal{J}'$ be an ideal in R/I. Define $J = \{\alpha \in R \mid \alpha + I \in J'\}$. First, we will show J is an ideal in R. Let $r \in R$ and $\alpha + I \in J'$. It follows that $\alpha \in J$. Notice that $(r + I)(\alpha + I) = r\alpha + I \in J'$ because J' is an ideal, so $r\alpha \in J$. It remains to be shown J is a ring.

Let $r, s \in J$. It follows that $r + I, s + I \in J'$. Because J' is a ring, we have

$$(r + I) - (s + I) = (r - s) + I \in J',$$

$$(r + I)(s + I) = rs + I \in J'.$$

Therefore, J is an ideal of R. Finally, we show $\phi(J) = J'$. Let $\alpha + I \in \phi(J)$. Since $\alpha \in J$, $\alpha + I \in J'$. Now let $\alpha + I \in J'$. Since $\alpha \in J$, $\alpha + I \in \phi(J)$. Therefore, ϕ is surjective.

Finally, we show ϕ is injective. Let $J_1, J_2 \in \mathcal{J}$ such that $\phi(J_1) = \phi(J_2)$. It follows that for all $\alpha \in J_1$, there exists $\beta \in J_2$ such that $\alpha + I = \beta + I$. By properties of cosets, $\alpha \in \beta + I$. Since J_2 is a subring of R, it's a group under addition, and is thus closed. Since $I \subseteq J_2$ and $\beta \in J_2$, it follows that $\beta + I \subseteq J_2$. Therefore, $\alpha \in J_2$, and thus $J_1 \subseteq J_2$. Applying the same logic in reverse shows $J_1 = J_2$. Therefore, ϕ is a bijection.

Part (b). Let R/M be a field. It follows that there exist precisely two ideals of R/M, namely R/M and $\{M\}$, since M is the additive identity. By part (a), there exist two ideals of R containing M. Suppose M is not the maximal ideal. It follows that there exists a maximal ideal M' such that $M \subset M' \subset R$ are all ideals. This implies there exist three ideals, which is a contradiction. Therefore, M is the maximal ideal.

Now we show the opposite. Let M be the maximal ideal. It follows that there are precisely two ideals containing M, namely M and R. Therefore, there exist two ideals in R/M. Since a nontrivial ring always contains two trivial ideals, we know R/M is nontrivial, and thus contains only the trivial ideals. By problem 4, R/M is a field.

Part (c). Let P be a prime ideal, and let $a + P, b + P \in R/P$. It follows that

$$(a + P)(b + P) = ab + P.$$

Notice that P is the additive identity in R/P. Suppose $a + P$ and $b + P$ are zero divisors; that is, $ab + P = P$. By properties of cosets, $ab + P = P$ if and only if $ab \in P$. Since P is prime, $ab \in P$ if and only if $a \in P$ or $b \in P$. Therefore, if $ab + P = P$, then by properties of cosets $a + P = P$ or $b + P = P$. Since $P = 0$ in R/P, there are no zero divisors in R/P. Since R is a commtutative ring with unity, R/P is an integral domain by proposition 1.

Now we show the converse. Let R/P be an integral domain. Let $a+P, b+P \in R/P$. Consider

$$(a+P)(b+P) = ab+P.$$

Since R/P is an integral domain, $ab+P = P$ if and only if either $a+P = P$ or $b+P = P$. By properties of cosets, this is equivalent to the statement that $ab \in P$ if and only if $a \in P$ or $b \in P$. Therefore, P is prime.

Part (d). Let M be the maximal ideal of R. By part (b), R/M is a field. Since fields are integral domains, by part (c), M is prime. However, there exist prime ideals that are not maximal. In any integral domain, $\{0\}$ is a prime ideal since no zero divisors exist. However, not all integral domains are fields. By problem 4, this means for any integral domain that is not a field, $\{0\}$ is prime, but not maximal.